



CHALLENGE 08 / 20

THE SIXTY-SECOND DATA LEAK

ULTRA-SHORT-DURATION EXFILTRATION SCOPING

NATIONAL LEVEL HACKATHON • 2026

POWERED BY SYNTHOQUEST PVT LTD

Student Project Development Challenge • No organizer-provided evidence package.

1 | PROBLEM BACKGROUND

An insurer with 80 million policies saw a 60-second outbound burst from a database replica to an external cloud address at 03:12:41. Traffic was encrypted with no inspection. Flow counters from the firewall, host and switch disagree: they report between 6 and 9 GB. The replica holds policy, claims, health and identity data; some columns are encrypted, some are tokenised and some are stored compressed.

Between 03:11 and 03:13 queries ran under an analytics service account, but query logging is sampled. Storage-level counters and snapshot activity also exist. The attacker may have read data through the database, through an export, or through the storage layer, bypassing database logging.

The regulators require the insurer to state which data categories left and approximately how many individuals are affected.

2 | PROBLEM STATEMENT

Develop a **Short-Burst Exfiltration Scoping System** that, from burst volume and timing and indirect host, database and storage traces, produces defensible lower and upper bounds on the data categories, record counts and individuals affected, and identifies the likely extraction mechanism.

The system must also quantify how much each additional piece of evidence would tighten those bounds.

Hard constraints

- No payload is visible and the counters disagree.
- Compression, encryption and tokenisation change the relationship between bytes and records.
- The extraction path may bypass database logging entirely.
- Bounds must be valid: in the team's own tests the true scope must fall inside the stated bounds at the stated rate.
- Counts must be produced per data category and per individual in a form usable in a regulatory notification.

No evidence package or dataset is provided. Teams decide what to build, what data they need, how to simulate the environment and how to validate the result. No tools or architecture are prescribed.

3 | MANDATORY FEATURES

Teams must develop a working software solution. The system must provide the following features. How each is achieved is for the team to decide.

01	Counter Reconciliation	Ability to reconcile disagreeing volume measurements.
02	Bytes-to-Records Modelling	Ability to relate bytes to records under compression and encryption.
03	Extraction Mechanism Inference	Ability to infer query, export, snapshot or storage-level extraction.
04	Bound Construction	Ability to produce valid lower and upper bounds.
05	Category and Individual Counting	Ability to count affected categories and individuals.
06	Logging-Bypass Reasoning	Ability to reason when database logs miss the extraction.
07	Evidence Value Analysis	Ability to quantify which new evidence would narrow the bounds.
08	Bound Coverage Validation	Ability to show that stated bounds hold at their stated rate.

The system must clearly distinguish between:

CERTAINLY AFFECTED	POSSIBLY AFFECTED	CERTAINLY UNAFFECTED	ESTIMATED	UNKNOWN
--------------------	-------------------	----------------------	-----------	---------

A data scope that cannot be adequately bounded must not be presented as precise.

Build, don't search. This is a development challenge. Teams must create the system and the controlled environment that proves it works. Submitting an analysis of a given dataset is not a solution.

4 | EXPECTED OUTPUT

Live demonstration. The team must show that its system can handle a complex controlled scenario created by the team. The demonstration must show:

01	Teams' own simulated database estate with several storage forms
02	Simulated extractions by different mechanisms
03	Reconciliation of disagreeing counters
04	Bounds on categories, records and individuals
05	Identification of the extraction mechanism
06	Measured coverage of the bounds
07	Evidence that would narrow the bounds

The team must be able to explain why the system reached each major conclusion.

Validation standard (every team)

- Build a scenario generator with known ground truth and evaluate on at least ten distinct scenarios, including ones with deliberately misleading evidence.
- At the demonstration, judges change scenario parameters on the spot. The system must adapt without being rebuilt.
- Include an adversary variant that actively tries to defeat the system, and report how the system held up.
- Stated confidence must match measured accuracy. Overconfident or unexplained conclusions lose marks.
- No real systems, real personal data or functional malware may be used.

Submission. Each team must submit:

01	Working Prototype	A functional implementation of the proposed solution.
02	Source Code	Complete source code required to reproduce the project.
03	Technical Architecture	Architecture and component description.
04	Validation Environment	A controlled environment created by the team to demonstrate the solution.
05	Technical Documentation	Implementation, design and testing documentation.
06	Forensic Investigation Report	A professional report generated from the demonstrated investigation.
07	Demonstration	Live technical demonstration to the judging panel.
08	Limitations	Known limitations and situations where the system cannot establish a reliable conclusion.